TFI-Report 23-000532-03

Reaction to fire test
for classification according to EN 13501-1:2018

Customer

ASB Systembau Horst Babinsky GmbH
Fabrikstraße 14
83371 Stein a. d. Traun

Product

Floor construction
ASB GlassFloor

This report includes 17 pages.

Aachen, 13.06.2023

Dr. Jens-Christian Winkler
Authorized Manager

The present document is provided with an advanced electronic signature.

This report only applies to the tested samples and has been established to the best of our knowledge. Only the entire report shall be reproduced. Under no circumstances, extracts shall be used. Furthermore, we apply the "General Terms and Conditions for the Execution of Contracts" of the TFI Aachen GmbH, also with regard to the order execution.

The test result does not include any addition or deduction for uncertainties due to measurement, sample preparation, sample collection and production tolerances.

1 Transaction

Order date	16.02.2023
Order number	23-000532 - AB2300406
Your reference	Projektnr.: 500400E/30910, Bauvorhaben: Entwicklung
Product designation	ASB GlassFloor
Charge	-
Item number	-
TFI sample number	2300765
Date of sample receipt	04.05.2023
Sampling performed by	Auftraggeber/Customer see Probenahmeprotokoll
Test period	23.05.2023 - 02.06.2023

Professionally responsible for the examinations of the fire department

Ulrike Balg
+49 241 9679133
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2 Product description

TFI Sample Number 2300765

Total thickness [mm]	11,31
Surface related mass [g/m²]	25872
Delivery form	In Probengröße angeliefert/Supplied in sample size

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USt-IdNr. DE209411312
Managing Director
Dr. Jacqueline Lemm

3 Test methods / requirements

Test order:

Testing according to EN ISO 11925-2:2020, Part 2 Surface [a]
Testing according to EN ISO 9239-1:2010 Part 1 [a]

a ... Die mit a gekennzeichneten Ergebnisse basieren auf nach EN ISO/IEC 17025 akkreditierten Prüfungen./The reports marked a are based on tests accredited in accordance with EN ISO/IEC 17025.

The following applies to test method EN ISO 9239-1:

Deviation	none**
Adhesion	none
Substrate according to EN 13238:2010	fibre cement board
Joint according to EN ISO 9239-1:2010	yes and none
Laboratory spray extraction cleaning procedure	nicht relevant
Conditioning	according to EN 13238:2010

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4 Results

Ignition of the specimen	no
150 mm mark reached	no
Ignition of the filter paper due to burning droplets	no
Critical heat flux without direction in kW / m²	11,0
Integrated smoke density without direction in % x min	13

Unless otherwise specified by the test standard, the measurement results are evaluated without taking into account the measurement uncertainty with regard to compliance with limit values.

The test results refer only to the behavior of the samples of a construction product under the particular conditions of the test; they are not to be understood as the sole criterion for evaluating the potential fire hazard of the construction product in the application.

**This test report serves as a basis for the preparation of a classification report according to EN 13501-1:2018.

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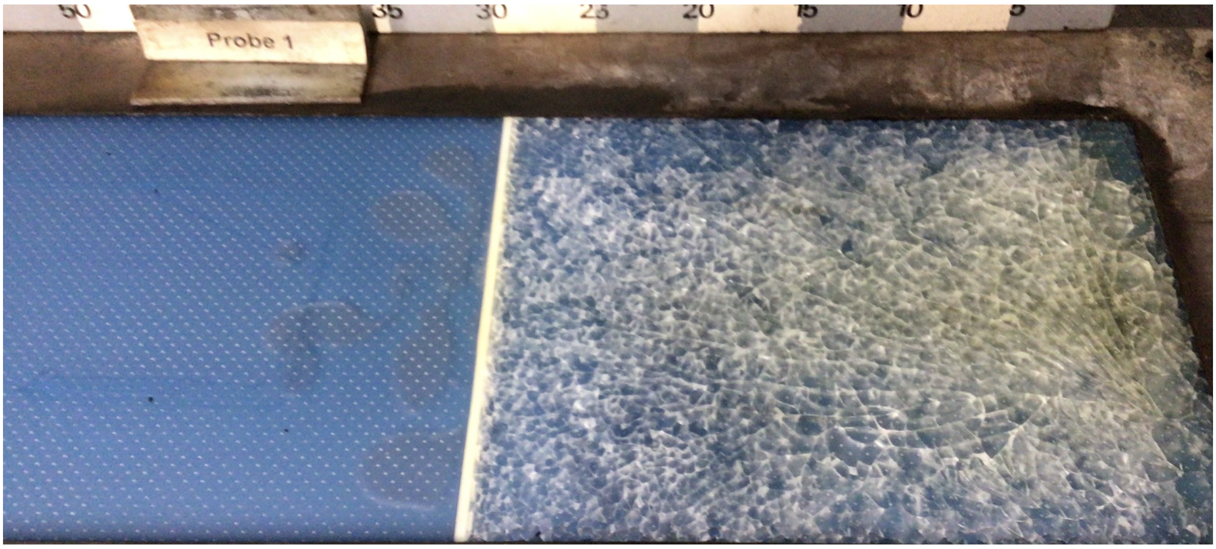
5 Pictures

5.1. Sample construction before testing

5.2. Reaction to fire sample 1 without direction



Notified Body
No. 1658

Testing laboratory,
inspection and certification
body recognised by the DIBt
(Deutsches Institut für Bautechnik)

DAkkS
Deutsche
Akkreditierungsstelle
D-PL-17152-01-00
Accredited for the methods indicated in
the partial reports to the DAkkS certificate

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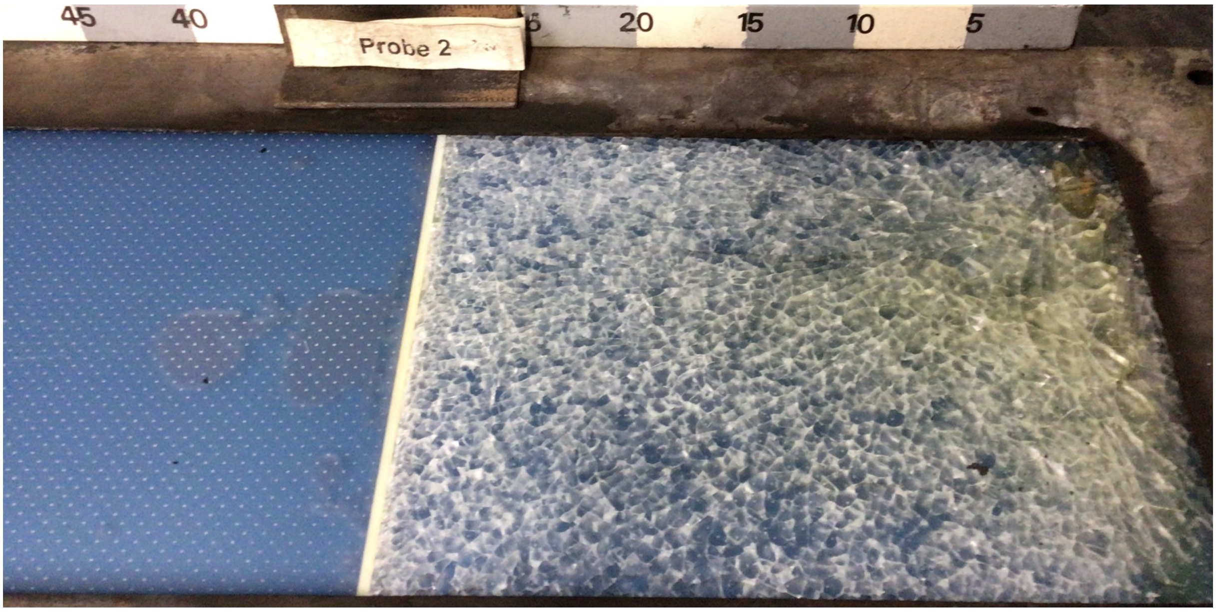
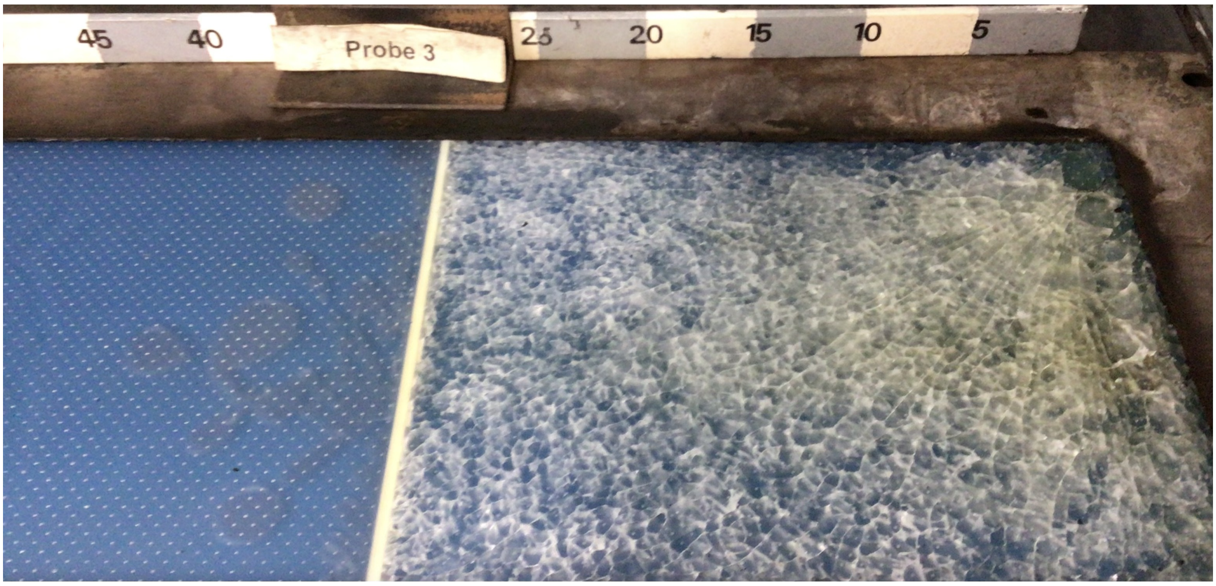
5.3. Reaction to fire sample 2 without direction



5.4. Brandverhalten Prüfprobe 3 without direction



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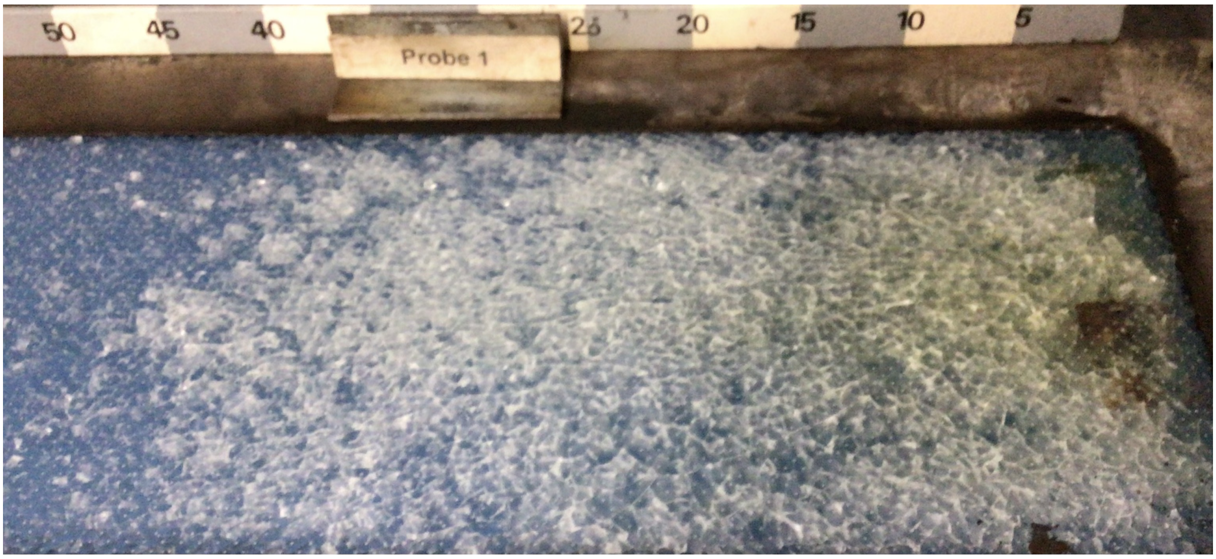
5.5. Reaction to fire sample 1 without direction without joint at a distance of 250 mm



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6 Partial reports

Testing according to EN ISO 11925-2:2020, Part 2 Surface [a]
Testing according to EN ISO 9239-1:2010 Part 1 [a]

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7 Annexes

Manufacturing report

Remark:
to point 2 Product description
The product description refers exclusively to the glass top (secruit etched glass) without substructure. The tested overall construction is shown under point 5.1 and described in the manufacturing protocol.

to point 4 Results
The mean value is formed from the set of results of the three test samples with joint (at a distance of 300 mm from the zero line of the sample).

Partial Report – Testing according to EN ISO 11925-2:2020, Part 2 Surface ignition

1 Test Method / Requirements

EN ISO 11925-2:2020, Part 2	Ignitability of products subjected to direct impingement of flame –Part 2: Single-flame source test
Substrate according to EN 13238:2010	Fibre cement board
Type of fixation	Loosely laid
Conditioning	according to EN 13238:2010
Conditioning	Surface ignition
Type of ignition	15
Deviation from the standard	-none-

2 Results

Parameter	Specimen no.					
	1	2	3	4	5	6
Orientation to the direction of production	without direction	without direction	without direction	without direction	without direction	without direction
Ignition of the specimen	no	no	no	no	no	no
Flame tip (moment [s])	< 150mm (n.r.)	< 150mm (n.r.)	< 150mm (n.r.)	< 150mm (n.r.)	< 150mm (n.r.)	< 150mm (n.r.)
Maximum flame height [mm] (moment [s])	n.r	n.r	n.r	n.r	n.r	n.r
Burning droplets	no	no	no	no	no	no
Ignition of the filter paper	no	no	no	no	no	no

Observations: -none-

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Partial Report – Testing according to EN ISO 9239-1:2010 Part 1

1 Results

Sample designation 2300765 with joint

Sample

Sample No.: 1

Direction: without direction

Observation

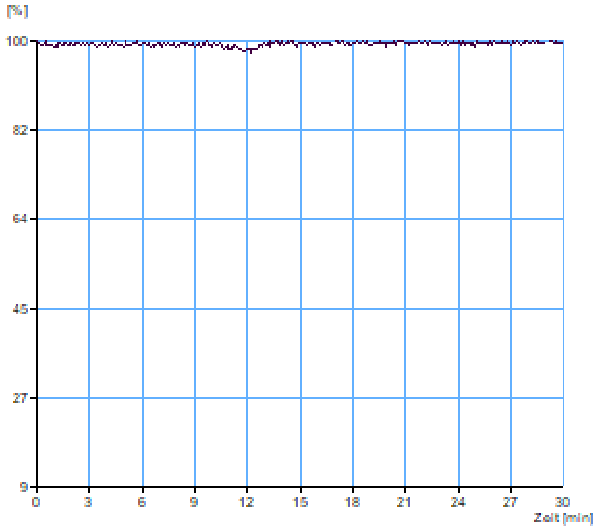
molten/singed during pre-radiation up to	0 mm
buckled/contracted from pilot flame area up to	0 mm
penetration of flame through substrate	-
transitory flaming	-
blistering	-
glowing, after flame has extinguished	-

Results

Position [mm]	Time [min:s]	Heat Flow [kW/m²]
50	-	-
100	-	-
150	-	-
200	-	-
250	-	-
300	-	-
350	-	-
400	-	-
450	-	-
500	-	-
550	-	-
600	-	-
650	-	-
700	-	-
750	-	-
800	-	-
850	-	-
900	-	-
950	-	-
1000	-	-

Time [min:s]	Position [mm]	Heat Flow [kW/m²]
10:00	-	-
20:00	-	-
30:00	-	-

Light transmission



CHF [kW/m²]	>= 11
HF_30 [kW/m²]	0.00
Smoke density integral [%*min]	17.9
Flame extinguished after [min:s]	00:00
max. burnt distance [mm]	0
max. light attantuation [%]	2.2

Sample designation	2300765 with joint
Sample	
Sample No.:	2
Direction:	without direction

Observation

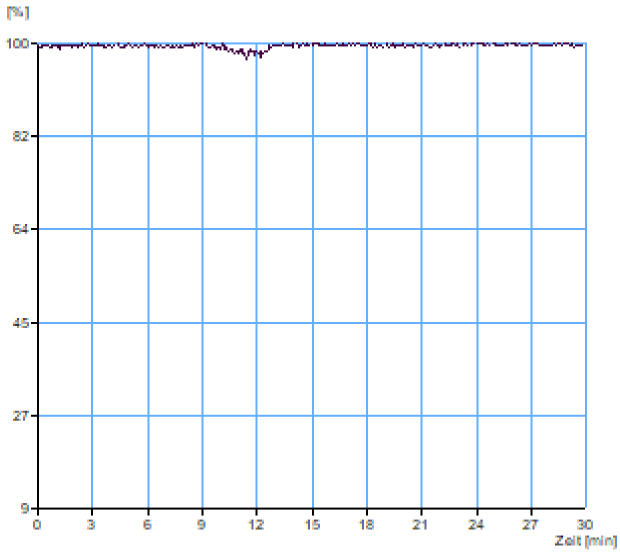
molten/singed during pre-radiation up to	0 mm
buckled/contracted from pilot flame area up to	0 mm
penetration of flame through substrate	-
transitory flaming	-
blistering	-
glowing, after flame has extinguished	-

Results

Position [mm]	Time [min:s]	Heat Flow [kW/m²]
50	-	-
100	-	-
150	-	-
200	-	-
250	-	-
300	-	-
350	-	-
400	-	-
450	-	-
500	-	-
550	-	-
600	-	-
650	-	-
700	-	-
750	-	-
800	-	-
850	-	-
900	-	-
950	-	-
1000	-	-

Time [min:s]	Position [mm]	Heat Flow [kW/m²]
10:00	-	-
20:00	-	-
30:00	-	-

Light transmission



CHF [kW/m²]	>= 11
HF_30 [kW/m²]	0.00
Smoke density integral [%*min]	15.4
Flame extinguished after [min:s]	12:00
max. burnt distance [mm]	0
max. light attantuation [%]	3.2

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Sample designation	2300765 with joint
Sample	
Sample No.:	3
Direction:	without direction

Observation

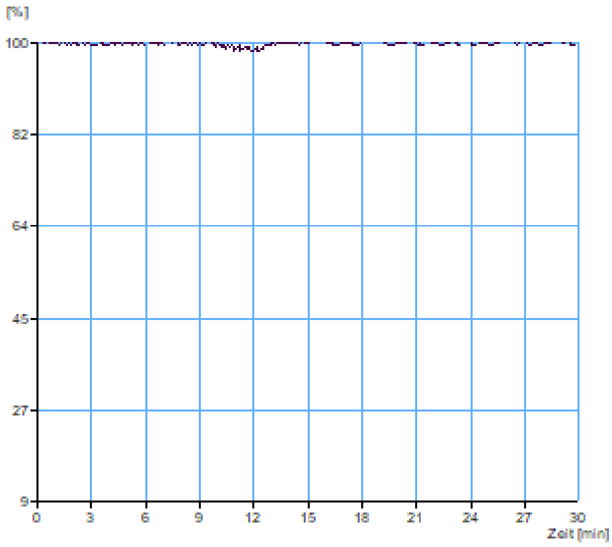
molten/singed during pre-radiation up to	0 mm
buckled/contracted from pilot flame area up to	0 mm
penetration of flame through substrate	-
transitory flaming	-
blistering	-
glowing, after flame has extinguished	-

Results

Position [mm]	Time [min:s]	Heat Flow [kW/m²]
50	-	-
100	-	-
150	-	-
200	-	-
250	-	-
300	-	-
350	-	-
400	-	-
450	-	-
500	-	-
550	-	-
600	-	-
650	-	-
700	-	-
750	-	-
800	-	-
850	-	-
900	-	-
950	-	-
1000	-	-

Time [min:s]	Position [mm]	Heat Flow [kW/m²]
10:00	-	-
20:00	-	-
30:00	-	-

Light transmission



CHF [kW/m²]	>= 11
HF_30 [kW/m²]	0.00
Smoke density integral [%*min]	7.1
Flame extinguished after [min:s]	12:00
max. burnt distance [mm]	0
max. light attantuation [%]	1.7

Sample designation 2300765 without joint

Sample

Sample No.: 1

Direction: without direction

Observation

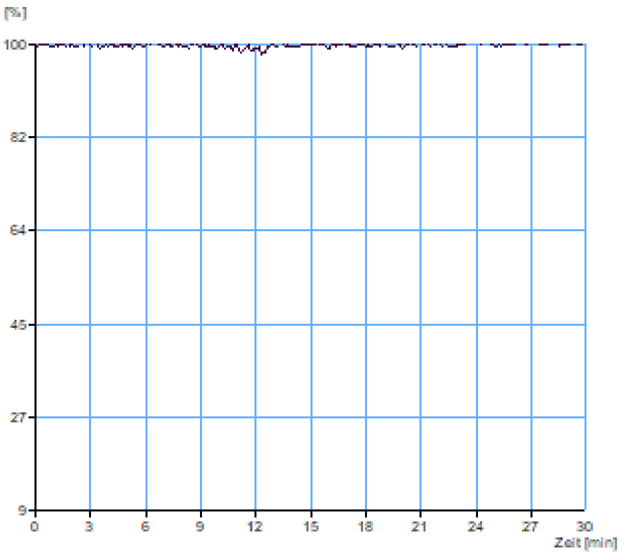
molten/singed during pre-radiation up to	0 mm
buckled/contracted from pilot flame area up to	0 mm
penetration of flame through substrate	-
transitory flaming	-
blistering	-
glowing, after flame has extinguished	-

Results

Position [mm]	Time [min:s]	Heat Flow [kW/m²]
50	-	-
100	-	-
150	-	-
200	-	-
250	-	-
300	-	-
350	-	-
400	-	-
450	-	-
500	-	-
550	-	-
600	-	-
650	-	-
700	-	-
750	-	-
800	-	-
850	-	-
900	-	-
950	-	-
1000	-	-

Time [min:s]	Position [mm]	Heat Flow [kW/m²]
10:00	-	-
20:00	-	-
30:00	-	-

Light transmission



CHF [kW/m²]	>= 11
HF_30 [kW/m²]	0.00
Smoke density integral [%*min]	7.6
Flame extinguished after [min:s]	12:00
max. burnt distance [mm]	0
max. light attantuation [%]	1.9

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Observations:

Test sample 1 with joint:

- no ignition
- after 1425 seconds shattering/ bursting of the glass pane up to the joint

Test sample 2 with joint:

- no ignition
- after 883 seconds shattering/ bursting of the glass pane up to the joint

Test sample 3 with joint:

- no ignition
- after 1702 seconds shattering/ bursting of the glass pane up to the joint

Test sample 1 without joint:

- no ignition
- after 1000 seconds shattering/ bursting of the glass pane

Test sample 1 – 3 with joint, test sample 1 without joint:

- melting the foil between the glass panes
- droplet from the test sample edge

2 Classification criteria according to EN 13501-1:2018, Table 2

Class	Test method(s)	Classification criteria	Additional classifications
B_{fl}	EN ISO 9239-1	Critical flux ≥ 8.0 kW / m²	Smoke production[1]
	EN ISO 11925-2, Exposure 15 s	Flame height ≤ 150 mm within 20 s	-
C_{fl}	EN ISO 9239-1	Critical flux ≥ 4.5 kW / m²	Smoke production[1]
	EN ISO 11925-2, Exposure 15 s	Flame height ≤ 150 mm within 20 s	-
D_{fl}	EN ISO 9239-1	Critical flux ≥ 3.0 kW / m²	Smoke production[1]
	EN ISO 11925-2, Exposure 15 s	Flame height ≤ 150 mm within 20 s	-
E_{fl}	EN ISO 11925-2, Exposure 15 s	Flame height ≤ 150 mm within 20 s	-

[1] s1 = smoke ≤ 750 % x min, s2 = smoke > 750 % x min

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Herstellprotokoll Systemaufbauten (Auftrag Nr.)

Prüflabor:	TFI Aachen GmbH	**Probennehmer:** (Organisation und Name des Probennehmers)	
Name des Herstellers / Händlers am Probenahmeort:		**Produkthersteller:** (falls abweichend vom Firmennamen am Probenahmeort)	

Produktname: *(bindend für Prüfbericht)*	ASB GlassFloor	**Artikel-Nr. / Gruppe, Serie:**	
Gruppe/Serie	☐ CE: 1658-CPR- ☐ DIBt: ☐ TÜV-Interior: 70 710	**Probeart:**	☐ Sportbodensystem punkt-/mischelastisch ☐ Sportbodensystem mit elastischer Schicht (flächen-/kombielastisch) ☐ Sportbodensystem mit elastischer Konstruktion (flächen-/kombielastisch) ☐
Chargen-Nr. bzw. Kennung der Probe:		**Produktionszeitraum der Probe:**	vom bis

Datum der Probenahme / Uhrzeit:				**Lagerort:**	
Probe gezogen:	☐ aus laufender Produktion ☐ aus Lagerbeständen ☐ aus Rückstellproben	**Art der Lagerung vor Entnahme:**	☐ offen ☐ verpackt	**Verpackungsmaterial:**	

Besonderheiten: (mögliche negative Einflüsse durch Emissionen am Probenahmeort, Unklarheiten, Fragen, etc.)	☐ **Entnahme als Rückstellprobe**	☐ Gasbetriebene Gabelstapler
Flammschutzmittel (Pflichtangabe):	Enthalten eine oder mehrere Komponenten ein Flammschutzmittel?	☒ nein ☐ ja

Vorgesehene Prüfungen:	☐ Emissionsprüfung nach DIBt (Erstprüfung)	☐ Emissionsprüfung nach DIBt (Fremdüberwachung)	
	☐ Bestimmung der Brandklasse (RP)	☐ RP red. Probenzahl	☐ Kleinbrenner (KB)
☐ TÜV-Interior Zulassungs-/Erstprüfung	☐ Überwachungsprüfung wie Zulassungs-/Erstprüfung	☐ Konstruktionsmerkmale	☐ Sonstiges:
☐ TÜV-Interior Überwachung Qualität	☐ TÜV Interior Überwachung Emission	☐ Standard	☐ Premium Vergabekriterien V

schematische Darstellung des Aufbaus (Pflichtangabe)	☐ liegt bei	☐ wird innerhalb von drei Tagen nachgereicht
Aufbauanleitung:	☐ liegt bei	☐ wird innerhalb von drei Tagen nachgereicht

Hiermit bestätigen die Unterzeichner die Richtigkeit der oben gemachten Angaben. Die Probe wurde eigenhändig gemäß Probenahmeanleitung ausgewählt, entnommen und verpackt.

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Tel +49 (0)8621 9874 - 0 • Fax +49 (0)8621 9874 - 20

Unterschrift Probennehmer (bei Probenahme durch Dritte)

Unterschrift Unternehmen

Prüfkörperaufbau:

- Nennung der verwendeten Materialen von unten beginnend, eine Zeile je verwendetem Material
- Probengröße # für Emissionsmessungen 305 x 420 mm # für Radiant-Panel-Test: 105 x 23 cm # für Small flame test: 9 x 25 cm
- **Achtung** bei **PUR-Beschichtungen** für Emissionsprüfungen: Die letzte Schicht muss im Labor der Prüfstelle aufgetragen werden!

* bei flüssigen oder pastösen Materialien

Materialbezeichnung	Artikelnummer	Produktspezifikation (abZ, GEV, Norm, ...)	Charge	Herstelldatum bzw. Mindesthaltbarkeit * **+** Art der Verpackung	Schichtdicke/ Auftragsmenge*	Mischungs-verhältnis*	Applikation mit *	Trocknungs-zeit*
Securit etched glass			EN12150		5,00mm			
PVB Folie			EN 12543		1,25 mm			
Securit clear floatglass			EN 12150		5,00 mm			
Farbschicht					keine Angaben			
Polyurethanschaumauflage					2,00 mm			
Glsauflage und LED- Kanal bestehend aus Aluminium					22,0 mm			
Aluminium-Schwingkonstruktion Obergurt					8,00 mm			
Aluminium-Schwingkonstruktion Untergurt					10,0 mm			
Füße höhenverstellbar					60,0 mm-125 mm			